Lesson 14: P-Values; Power and Sample Size Determination

BSTA 551: Statistical Inference

Jessica Minnier

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Review: Where We Left Off

Lesson 13: Tests About a Population Mean and Proportion

- One-sample z test (σ known) and t test (σ unknown)
- Score test and exact binomial test for proportions
- Seven-step hypothesis testing procedure
- CI-test duality: a $100(1-\alpha)\%$ CI contains values not rejected at level α

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Today's Goals:

1. Understand P-values as a continuous measure of evidence against H_0
2. Prove that P-values are Uniform(0,1) under H_0 using the probability integral transformation
3. Understand conditional P-values and Fisher's exact test for categorical data
4. Use the noncentral t distribution for exact power calculations
5. Determine sample sizes for planning clinical trials using R
6. Apply multiple comparison adjustments (Bonferroni, Benjamini-Hochberg)

Roadmap for Today

Topic	Key Idea	Textbook
P-values	Probability of data as extreme or more extreme than observed	Devore 9.4
P-value distribution	Uniform under H_0; concentrated near 0 under H_a	Devore 9.4
Fisher's exact test	Conditional P-values eliminate nuisance parameters	—
Power for the t test	Noncentral t distribution	Devore 9.2
Sample size determination	`power.t.test()` in R	Devore 9.2
Multiple testing	FWER (Bonferroni) and FDR (Benjamini-Hochberg)	—

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Why This Matters

In medical research, P-values appear in virtually every published paper. Understanding what they mean — and what they do **not** mean — is essential for interpreting clinical evidence.

P-Values (Devore 9.4)

From Rejection Regions to P-Values

Recall the rejection region approach:

1. Fix α (e.g., 0.05)
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Two scenarios with the same decision:

- $\alpha = 0.05$, P-value = 0.049 $\Rightarrow$ barely reject H_0
- $\alpha = 0.05$, P-value = 0.00001 $\Rightarrow$ overwhelming evidence against H_0

The rejection region approach treats these identically!

Definition of the P-Value

Definition (Devore 9.4)

The **P-value** is the probability, calculated assuming that the null hypothesis is true, of obtaining a value of the test statistic at least as contradictory to H_0 as the value calculated from the available sample.

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Key points:

- The P-value **is** a probability
- It is calculated **assuming H_0 is true**
- It is a function of the **sample data** (and therefore a random variable!)
- We need to determine which values are "at least as contradictory" to H_0

Formal Definition of the P-Value

Let $T = T(X_1, \ldots, X_n)$ be a test statistic and let t_{obs} be its observed value. For a test of $H_0 : \theta = \theta_0$ vs H_a:

Definition (Formal)

The **P-value** is defined as:

$$\text{P-value} = \begin{cases} P_{\theta_0}(T \geq t_{obs}) & \text{upper-tailed test } (H_a : \theta > \theta_0) \\ P_{\theta_0}(T \leq t_{obs}) & \text{lower-tailed test } (H_a : \theta < \theta_0) \\ P_{\theta_0}(|T| \geq |t_{obs}|) & \text{two-tailed test } (H_a : \theta \neq \theta_0) \end{cases}$$

where $P_{\theta_0}(\cdot)$ denotes probability calculated under H_0.

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where $P_{\theta_0}(\cdot)$ denotes probability calculated under H_0.

Interpretation: The P-value measures how "extreme" or "surprising" the observed data would be if H_0 were true. Smaller P-values indicate data that is less compatible with H_0.

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Why "at least as extreme"?

If t_{obs} is unlikely under H_0, then values *even more extreme* than t_{obs} must also be unlikely. The P-value captures the total probability of all outcomes at least as unfavorable to H_0.

The Probability Integral Transformation

A fundamental result from probability theory underlies the distribution of P-values.

Theorem (Probability Integral Transformation)

Let X be a continuous random variable with CDF F_X. Then the random variable

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Proof: For $u \in (0, 1)$:

$$P(U \leq u) = P(F_X(X) \leq u) = P(X \leq F_X^{-1}(u)) = F_X(F_X^{-1}(u)) = u$$

The second equality uses the fact that F_X is monotone increasing (for continuous X). Since $P(U \leq u) = u$ for all $u \in (0, 1)$, we have $U \sim \text{Uniform}(0, 1)$. ■

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This is easily verified: $P(1 - U \leq v) = P(U \geq 1 - v) = 1 - (1 - v) = v$. ■

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Why this matters: For an upper-tailed test, the P-value is $1 - F_T(T)$, which is exactly the survival function evaluated at the test statistic!

P-Values Are Uniform Under H_0

We can now state and prove the central result:

Theorem (Distribution of the P-Value Under H_0)

Let T be a test statistic with continuous CDF F_T under H_0. For an upper-tailed test, define the P-value as

$$P = 1 - F_T(T)$$

Then under H_0, the P-value satisfies $P \sim \mathrm{Uniform}(0, 1)$.

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Proof: Under H_0, the test statistic T has CDF F_T. By the probability integral transformation, $F_T(T) \sim \mathrm{Uniform}(0, 1)$. By the corollary, $P = 1 - F_T(T) \sim \mathrm{Uniform}(0, 1)$. ■

Consequence: P-Value Tests Maintain the Correct Size

This uniformity result has a crucial consequence for hypothesis testing.

Definition: Size of a Test

The **size** of a hypothesis test is the probability of rejecting H_0 when H_0 is true—i.e., the Type I error rate:

$$\text{Size} = P_{H_0}(\text{Reject } H_0) = P(\text{Type I Error})$$

A test has **exact size** α if this probability equals α exactly. A test has size **at most** α if this probability is $\leq \alpha$.

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Theorem (Size of P-Value Tests)

If the P-value P has a $\text{Uniform}(0, 1)$ distribution under H_0, then the test that rejects H_0 when $P \leq \alpha$ has **exact size** α.

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Key insight: The P-value approach **automatically** guarantees that rejecting when P-value $\leq \alpha$ produces a test with Type I error rate exactly equal to α. This is not a coincidence—it's a direct consequence of the probability integral transformation!

Why This Matters: Equivalence of Approaches

Rejection region approach:

- Choose α, find critical value c such that $P_{H_0}(T \geq c) = \alpha$
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$$T \geq c \iff 1 - F_T(T) \leq 1 - F_T(c) = \alpha \iff \text{P-value} \leq \alpha$$

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The P-value encodes *all possible rejection decisions* for any choice of α:

- If P-value $= 0.03$: reject at $\alpha = 0.05$ but not at $\alpha = 0.01$
- If P-value $= 0.008$: reject at both $\alpha = 0.05$ and $\alpha = 0.01$

P-Values for Discrete Test Statistics

Technical Note

The probability integral transformation requires T to be **continuous**. For discrete test statistics (e.g., the binomial), the P-value may not be exactly Uniform $(0, 1)$.

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Consequence: For discrete tests, $P(P \leq \alpha) \leq \alpha$, with equality only when α happens to coincide with one of the achievable P-values. The test is **conservative** (actual Type I error $<$ nominal α).

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Resolution: Randomized tests can achieve exact size α, but are rarely used in practice. Instead, we accept the slight conservatism.

Conditional P-Values and Fisher's Exact Test

What is Fisher's Exact Test?

Fisher's exact test is a statistical test used to determine whether there is a significant association between two categorical variables in a 2×2 contingency table.

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Common applications in medical research:

- Is a genetic variant associated with disease risk?
- Does a treatment affect the probability of an adverse event?
- Is exposure to a risk factor associated with an outcome?
- Does a diagnostic test result predict disease status?

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Why "Exact"?

Unlike the chi-squared test (which relies on large-sample approximations), Fisher's exact test computes the **exact** P-value by enumerating all possible tables. This makes it particularly valuable for small samples or sparse tables where asymptotic approximations may be unreliable.

Fisher's Exact Test: The Setup

Consider data from a case-control or cross-sectional study:

	Outcome+	**Outcome−**	**Total**
Group A	a	b	a + b
Group B	c	d	c + d
Total	a + c	b + d	n

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The question: Is the proportion with Outcome+ different between Group A and Group B?

Equivalently: Are group membership and outcome **independent** or **associated**?

- H_0: No association (independence): $P(\text{Outcome+} \mid \text{Group A}) = P(\text{Outcome+} \mid \text{Group B})$
- H_a: There is an association: $P(\text{Outcome+} \mid \text{Group A}) \neq P(\text{Outcome+} \mid \text{Group B})$

Fisher's Exact Test vs Chi-Squared Test

Feature	Fisher's Exact Test	Chi-Squared Test
P-value	Exact	Approximate (asymptotic)
Sample size	Valid for any n	Requires "large" n (expected counts ≥ 5)
Computation	Enumerates all tables	Simple formula
Sparse tables	Handles well	May be unreliable
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When to Use Fisher's Exact Test

- Any expected cell count < 5
- Total sample size < 20-30
- When exact P-values are required (regulatory submissions)
- One-sided hypotheses about association direction

For large samples with adequate expected counts, chi-squared and Fisher's give nearly identical results.

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Testing H_0: "Group and outcome are independent" does **not** tell us:

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- What proportion have the outcome

These **nuisance parameters** affect the distribution of any test statistic, yet we don't know their values!

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These **nuisance parameters** affect the distribution of any test statistic, yet we don't know their values!

Fisher's solution: Condition on the sufficient statistics for the nuisance parameters, eliminating them from the problem entirely.

The Problem of Nuisance Parameters

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Example: Testing independence in a 2×2 table

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The problem: Under H_0, the marginal probabilities $p_1 = P(\text{Exposed})$ and $p_2 = P(\text{Disease+})$ are **nuisance parameters**—unknown quantities not of direct interest.

Why Nuisance Parameters Complicate P-Values

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Fisher's exact test takes approach (2), which yields an **exact** P-value that is valid for any sample size.

The Conditionality Principle

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For nuisance parameters: If S is a sufficient statistic for the nuisance parameter η, then conditioning on S eliminates the dependence on η.

Fisher's Exact Test: Setup

Consider a 2×2 table with fixed total n. Let:

- X = number in cell (1,1) (Exposed and Disease+)
- Row totals $r_1 = a + b$ and $r_2 = c + d$ (fixed by design or conditioning)
- Column totals $c_1 = a + c$ and $c_2 = b + d$

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Under H_0 (independence) with both margins fixed, the cell count X follows a **hypergeometric distribution**:

$$P(X = x \mid r_1, r_2, c_1, c_2) = \frac{\binom{c_1}{x}\binom{c_2}{r_1 - x}}{\binom{n}{r_1}}$$

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Key observation: This distribution has **no unknown parameters**! The nuisance parameters have been eliminated.

Why Condition on the Margins?

Theorem (Sufficiency of Margins)

Under H_0 (independence), the row and column marginal totals (r_1, c_1) are **jointly sufficient** for the nuisance parameters (p_1, p_2).

Why Condition on the Margins?

Theorem (Sufficiency of Margins)

Under H_0 (independence), the row and column marginal totals (r_1, c_1) are **jointly sufficient** for the nuisance parameters (p_1, p_2).

Proof sketch: Under independence, the joint probability of a 2×2 table is:

$$P(a, b, c, d) = \frac{n!}{a!b!c!d!} p_1^{r_1} (1 - p_1)^{r_2} p_2^{c_1} (1 - p_2)^{c_2}$$

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By the factorization theorem, since the likelihood factors into a term depending on (p_1, p_2) only through (r_1, c_1), the margins are sufficient for the nuisance parameters.

The Conditional Distribution Eliminates Nuisance Parameters

Conditioning on the sufficient statistic (r_1, c_1):

$$P(X = x \mid r_1, c_1) = \frac{P(X = x, r_1, c_1)}{P(r_1, c_1)}$$

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After simplification (the nuisance parameters cancel!):

$$P(X = x \mid r_1, c_1) = \frac{\binom{c_1}{x}\binom{n-c_1}{r_1-x}}{\binom{n}{r_1}}$$

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This is the hypergeometric distribution with parameters n, r_1, and c_1—all **known** from the observed data!

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This is the hypergeometric distribution with parameters n, r_1, and c_1—all **known** from the observed data!

Key Insight

By conditioning on the margins, we transform a problem with nuisance parameters into one where the null distribution is completely specified.

Fisher's Exact Test: The Conditional P-Value

The **conditional P-value** is computed using the hypergeometric distribution:

$$\text{P-value} = P(X \geq x_{obs} \mid r_1, c_1) = \sum_{x=x_{obs}}^{\min(r_1, c_1)} \frac{\binom{c_1}{x}\binom{n-c_1}{r_1-x}}{\binom{n}{r_1}}$$

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(For a two-sided test, we sum probabilities for outcomes as extreme or more extreme in either direction.)

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(For a two-sided test, we sum probabilities for outcomes as extreme or more extreme in either direction.)

Properties:

- **Exact:** Valid for any sample size, including small samples
- **Free of nuisance parameters:** No need to estimate marginal probabilities
- **Discrete:** P-value can only take finitely many values (test is slightly conservative)

Example: Fisher's Exact Test in R

Setting: A clinical trial investigates whether a new treatment reduces adverse events.

```r
# Observed 2x2 table
#                Adverse Event   No Event
# Treatment            2             18        (20)
# Placebo              8             12        (20)
#                     (10)          (30)       (40)

adverse_table <- matrix(c(2, 8, 18, 12), nrow = 2, byrow = TRUE,
                        dimnames = list(Group = c("Treatment", "Placebo"),
                                        Outcome = c("Adverse", "None")))
adverse_table
```

```
           Outcome
Group       Adverse None
  Treatment       2    8
  Placebo        18   12
```

Example: Fisher's Exact Test Output

```r
# Fisher's exact test
fisher_result <- fisher.test(adverse_table)
fisher_result
```

```
	Fisher's Exact Test for Count Data

data:  adverse_table
p-value = 0.06483
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.01546086 1.08434068
sample estimates:
odds ratio
 0.1743672
```

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95 percent confidence interval:
 0.01546086 1.08434068
sample estimates:
odds ratio
 0.1743672
```

```r
cat("P-value:", round(fisher_result$p.value, 4), "\n")
```

```
P-value: 0.0648
```

```r
cat("Odds ratio estimate:", round(fisher_result$estimate, 3), "\n")
```

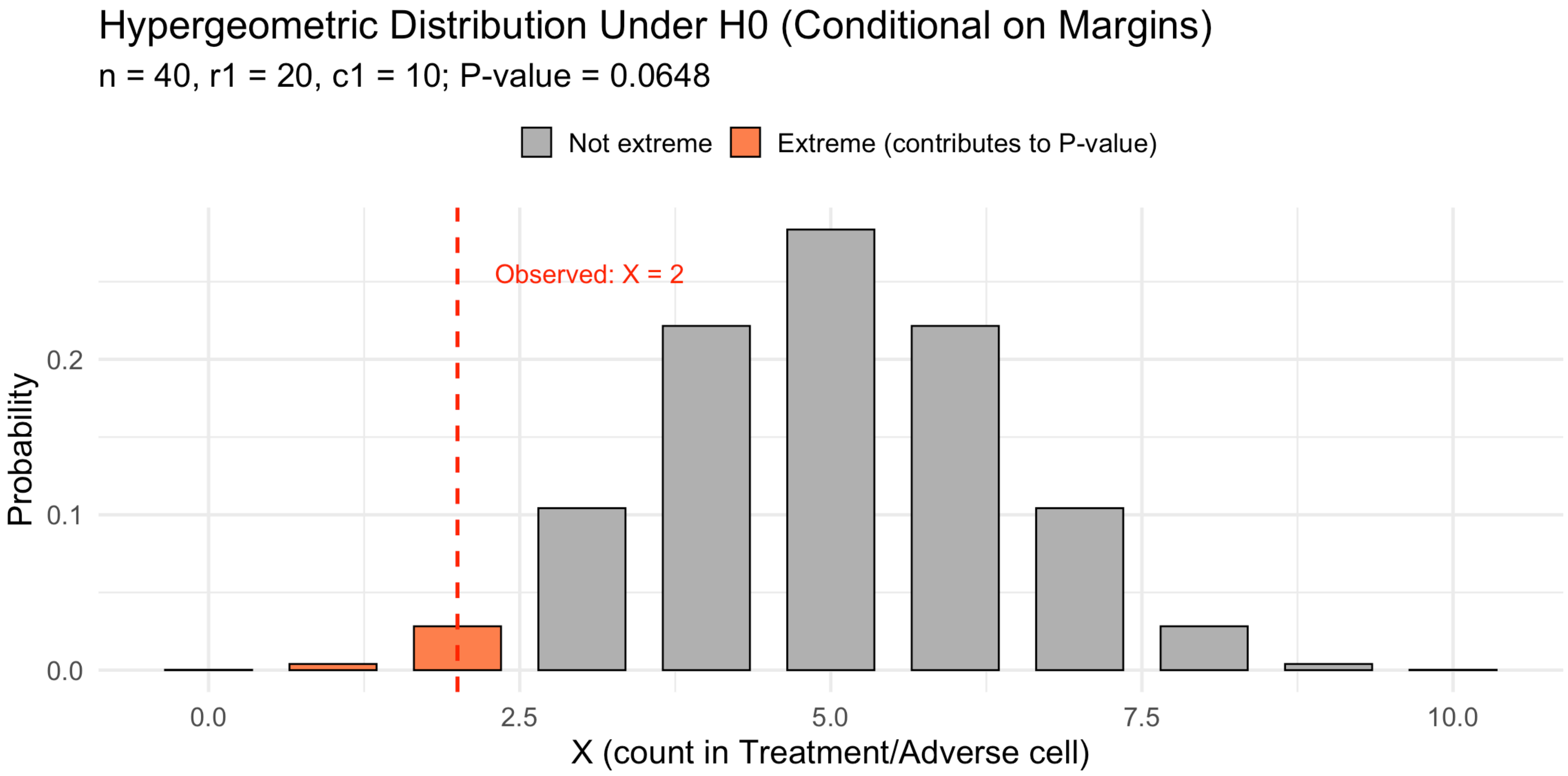
```
Odds ratio estimate: 0.174
```

```r
cat("95% CI for odds ratio:", round(fisher_result$conf.int[1], 3), "to",
    round(fisher_result$conf.int[2], 3), "\n")
```

```
95% CI for odds ratio: 0.015 to 1.084
```

Visualizing the Hypergeometric Distribution

▶ Code



Why Conditioning? A Deeper Look

The philosophical justification:

1. **Sufficiency Principle:** Inference should depend on the data only through sufficient statistics
2. **Conditionality Principle:** Inference should condition on ancillary statistics

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When combined, these principles suggest conditioning on sufficient statistics for nuisance parameters.

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When combined, these principles suggest conditioning on sufficient statistics for nuisance parameters.

The practical justification:

- Conditioning produces a test with **exact** size α
- No large-sample approximations needed
- Valid even for sparse tables or small samples

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1. **Sufficiency Principle:** Inference should depend on the data only through sufficient statistics
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When combined, these principles suggest conditioning on sufficient statistics for nuisance parameters.

The practical justification:

- Conditioning produces a test with **exact** size α
- No large-sample approximations needed
- Valid even for sparse tables or small samples

> ⚠ **Caveat**
>
> Conditioning can reduce power compared to unconditional tests. This is the price of exactness. For large samples, the chi-squared test (which estimates nuisance parameters) may be more powerful.

Connection to Randomization Tests

Fisher's exact test can also be viewed as a **randomization test**:

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Setup: Fix the margins (total exposed, total diseased). Under H_0, each possible table consistent with these margins is equally likely given the hypergeometric distribution.

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Setup: Fix the margins (total exposed, total diseased). Under H_0, each possible table consistent with these margins is equally likely given the hypergeometric distribution.

The P-value is the proportion of possible tables that show an association at least as extreme as observed.

Connection to Randomization Tests

Fisher's exact test can also be viewed as a **randomization test**:

Setup: Fix the margins (total exposed, total diseased). Under H_0, each possible table consistent with these margins is equally likely given the hypergeometric distribution.

The P-value is the proportion of possible tables that show an association at least as extreme as observed.

This connects to permutation testing: Under H_0, "Outcome" labels are exchangeable among subjects. The conditional P-value equals the proportion of permutations yielding a test statistic as extreme as observed.

```r
# Verify: Fisher's P-value via permutation simulation
set.seed(551)
n_perm <- 10000

# Reconstruct individual-level data matching adverse_table:
# Treatment: 2 adverse, 18 none; Placebo: 8 adverse, 12 none
treatment <- c(rep(1, 20), rep(0, 20))  # 20 treatment, 20 placebo
adverse <- c(rep(1, 2), rep(0, 18),     # Treatment: 2 adverse, 18 not
             rep(1, 8), rep(0, 12))      # Placebo: 8 adverse, 12 not

# Observed statistic: treatment patients with adverse events
obs_stat <- sum(treatment * adverse)  # = 2
cat("Observed statistic (treatment + adverse):", obs_stat, "\n")
```

```
Observed statistic (treatment + adverse): 2
```

Fisher's Exact Test: Complete R Workflow

Clinical scenario: A pilot study investigates whether a genetic variant is associated with response to a targeted cancer therapy.

```r
# Create the contingency table from raw data
patients <- tibble(
  patient_id = 1:45,
  variant = c(rep("Present", 18), rep("Absent", 27)),
  response = c(
    # Variant present: 12 responders, 6 non-responders
    rep("Responder", 12), rep("Non-responder", 6),
    # Variant absent: 8 responders, 19 non-responders
    rep("Responder", 8), rep("Non-responder", 19)
  )
)

# Cross-tabulation
response_table <- table(patients$variant, patients$response)
response_table
```

```
         Non-responder Responder
 Absent             19         8
 Present             6        12
```

Fisher's Exact Test: Analysis

```r
# Fisher's exact test (two-sided by default)
fisher_result <- fisher.test(response_table)

# Display full output
fisher_result
```

```
    Fisher's Exact Test for Count Data

data:  response_table
p-value = 0.03071
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
  1.121584 20.940769
sample estimates:
odds ratio
  4.570771
```

Fisher's Exact Test: Extracting Results

```r
# Extract components for reporting
cat("=== Fisher's Exact Test Results ===\n\n")
```

```
=== Fisher's Exact Test Results ===
```

```r
cat("P-value:", format(fisher_result$p.value, digits = 4), "\n")
```

```
P-value: 0.03071
```

```r
cat("Odds Ratio:", round(fisher_result$estimate, 2), "\n")
```

```
Odds Ratio: 4.57
```

```r
cat("95% CI for OR: [", round(fisher_result$conf.int[1], 2), ", ",
    round(fisher_result$conf.int[2], 2), "]\n\n")
```

```
95% CI for OR: [ 1.12 ,  20.94 ]
```

```r
# Decision at alpha = 0.05
alpha <- 0.05
if (fisher_result$p.value <= alpha) {
  cat("Decision: Reject H0 at alpha =", alpha, "\n")
  cat("Conclusion: There is a statistically significant association\n")
  cat("between the genetic variant and treatment response.\n")
} else {
  cat("Decision: Fail to reject H0 at alpha =", alpha, "\n")
  cat("Conclusion: Insufficient evidence of association.\n")
}
```

Fisher's Exact Test: One-Sided Alternatives

```r
# If we have a directional hypothesis (variant improves response)
fisher_greater <- fisher.test(response_table, alternative = "greater")
fisher_less <- fisher.test(response_table, alternative = "less")

cat("Two-sided P-value:", round(fisher_result$p.value, 4), "\n")
```

```
Two-sided P-value: 0.0307
```

```r
cat("One-sided (OR > 1):", round(fisher_greater$p.value, 4), "\n")
```

```
One-sided (OR > 1): 0.0157
```

```r
cat("One-sided (OR < 1):", round(fisher_less$p.value, 4), "\n")
```

```
One-sided (OR < 1): 0.9973
```

Fisher's Exact Test: One-Sided Alternatives

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# If we have a directional hypothesis (variant improves response)
fisher_greater <- fisher.test(response_table, alternative = "greater")
fisher_less <- fisher.test(response_table, alternative = "less")

cat("Two-sided P-value:", round(fisher_result$p.value, 4), "\n")
```

```
Two-sided P-value: 0.0307
```

```r
cat("One-sided (OR > 1):", round(fisher_greater$p.value, 4), "\n")
```

```
One-sided (OR > 1): 0.0157
```

```r
cat("One-sided (OR < 1):", round(fisher_less$p.value, 4), "\n")
```

```
One-sided (OR < 1): 0.9973
```

Choosing One-Sided vs Two-Sided

Use a **one-sided** test only when you have a strong *a priori* hypothesis about the direction of association, specified **before** seeing the data. In most exploratory studies, two-sided tests are more appropriate.

Comparing Fisher's Exact Test to Chi-Squared Test

```r
# Chi-squared test (asymptotic approximation)
chisq_result <- chisq.test(response_table)
chisq_result
```

```
	Pearson's Chi-squared test with Yates' continuity correction

data:  response_table
X-squared = 4.5938, df = 1, p-value = 0.03209
```

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data:  response_table
X-squared = 4.5938, df = 1, p-value = 0.03209
```

```r
# Compare P-values
cat("\nComparison of P-values:\n")
```

```
Comparison of P-values:
```

```r
cat("Fisher's exact test:", round(fisher_result$p.value, 4), "\n")
```

```
Fisher's exact test: 0.0307
```

```r
cat("Chi-squared test:    ", round(chisq_result$p.value, 4), "\n")
```

```
Chi-squared test:    0.0321
```

```r
cat("\nExpected counts (chi-squared assumes these are >= 5):\n")
```

```
Expected counts (chi-squared assumes these are >= 5):
```

```r
round(chisq_result$expected, 1)
```

Comparing Fisher's Exact Test to Chi-Squared Test

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Comparison of P-values:
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cat("Fisher's exact test:", round(fisher_result$p.value, 4), "\n")
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```
Fisher's exact test: 0.0307
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```
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```r
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```
Expected counts (chi-squared assumes these are >= 5):
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round(chisq_result$expected, 1)
```

Decision Rule Based on the P-Value

P-Value Decision Rule (Devore 9.4)

Select a significance level α. Then:

$$\text{Reject } H_0 \text{ if P-value} \leq \alpha; \qquad \text{Do not reject } H_0 \text{ if P-value} > \alpha$$

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Equivalence to the rejection region method:

- If the test statistic falls in the rejection region, then P-value $\leq \alpha$
- If the test statistic falls outside the rejection region, then P-value $> \alpha$

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Equivalence to the rejection region method:

- If the test statistic falls in the rejection region, then P-value $\leq \alpha$
- If the test statistic falls outside the rejection region, then P-value $> \alpha$

Advantage: The P-value encodes more information than a binary reject/fail-to-reject decision. It conveys the **strength** of evidence against H_0.

P-Values for z Tests

For a z test (test statistic Z with $Z \sim N(0,1)$ under H_0):

$$\text{P-value} = \begin{cases} 1 - \Phi(z) & \text{for an upper-tailed test } (H_a : \mu > \mu_0) \\ \Phi(z) & \text{for a lower-tailed test } (H_a : \mu < \mu_0) \\ 2[1 - \Phi(|z|)] & \text{for a two-tailed test } (H_a : \mu \neq \mu_0) \end{cases}$$

where $\Phi(z)$ is the standard normal CDF and z is the observed test statistic.

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where $\Phi(z)$ is the standard normal CDF and z is the observed test statistic.

```r
# Quick R reminders for P-value computation
z_obs <- 2.15
cat("Upper-tailed P-value:", 1 - pnorm(z_obs), "\n")
```

```
Upper-tailed P-value: 0.01577761
```

```r
cat("Lower-tailed P-value:", pnorm(z_obs), "\n")
```

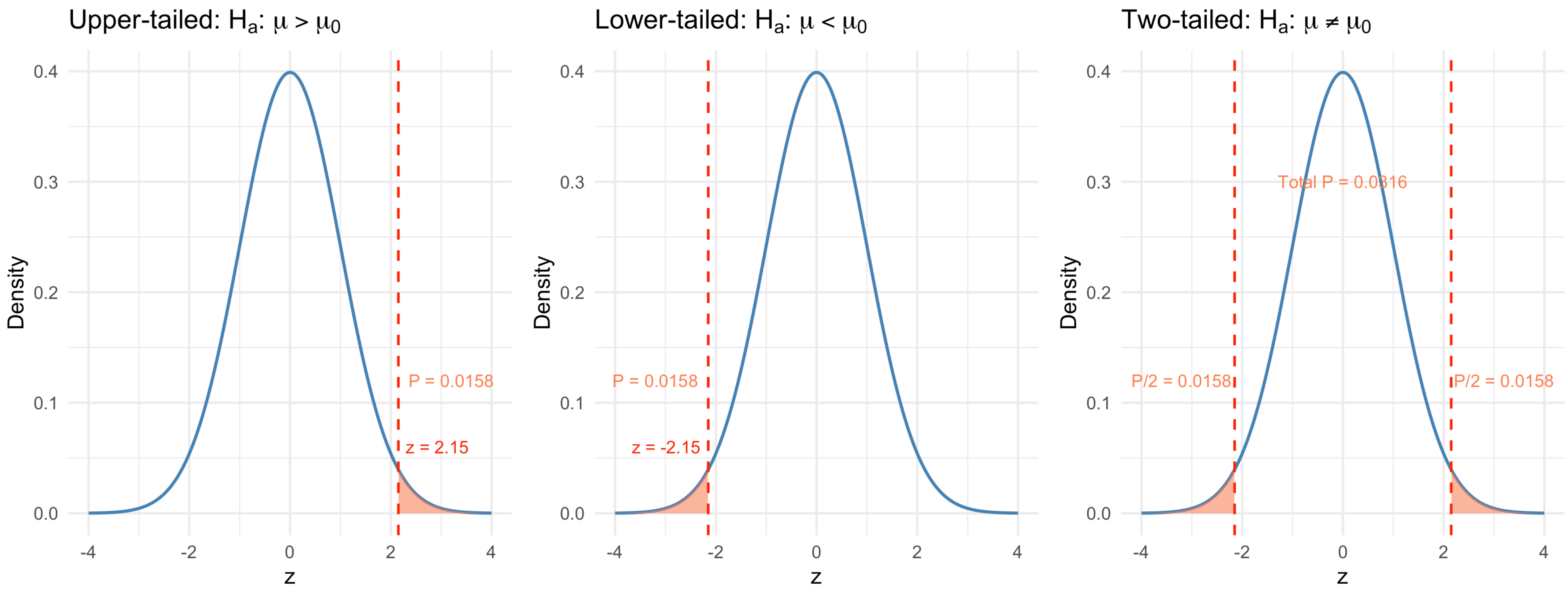
```
Lower-tailed P-value: 0.9842224
```

```r
cat("Two-tailed P-value:  ", 2 * (1 - pnorm(abs(z_obs))), "\n")
```

```
Two-tailed P-value:   0.03155521
```

Visualizing P-Values for z Tests

▶ Code



P-Values for t Tests

For the one-sample t test with $n-1$ degrees of freedom, the P-value formulas are analogous:

$$\text{P-value} = \begin{cases} P(T_{n-1} \geq t) & \text{upper-tailed} \\ P(T_{n-1} \leq t) & \text{lower-tailed} \\ 2 \cdot P(T_{n-1} \geq |t|) & \text{two-tailed} \end{cases}$$

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```r
# P-values for t tests in R
t_obs <- 2.15
df <- 14

cat("Upper-tailed P-value:", pt(t_obs, df = df, lower.tail = FALSE), "\n")
```

```
Upper-tailed P-value: 0.0247585
```

```r
cat("Lower-tailed P-value:", pt(t_obs, df = df, lower.tail = TRUE), "\n")
```

```
Lower-tailed P-value: 0.9752415
```

```r
cat("Two-tailed P-value:  ", 2 * pt(abs(t_obs), df = df, lower.tail = FALSE), "\n")
```

```
Two-tailed P-value:   0.04951701
```

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```r
# P-values for t tests in R
t_obs <- 2.15
df <- 14

cat("Upper-tailed P-value:", pt(t_obs, df = df, lower.tail = FALSE), "\n")
```

```
Upper-tailed P-value: 0.0247585
```

```r
cat("Lower-tailed P-value:", pt(t_obs, df = df, lower.tail = TRUE), "\n")
```

```
Lower-tailed P-value: 0.9752415
```

```r
cat("Two-tailed P-value:  ", 2 * pt(abs(t_obs), df = df, lower.tail = FALSE), "\n")
```

```
Two-tailed P-value:   0.04951701
```

R Functions

`t.test()` and `prop.test()` automatically compute P-values. Manual computation is needed only for understanding and for non-standard situations.

Medical Examples: Computing P-Values

Example: Hemoglobin A1c Levels in Diabetic Patients

Setting: A diabetes clinic is evaluating whether a new management protocol changes mean hemoglobin A1c (HbA1c) levels. The standard target is 7.0%. A sample of $n = 20$ patients on the new protocol gives $\bar{x} = 7.42\%$ and $s = 0.95\%$.

Question: Is there evidence that the new protocol produces different mean HbA1c than the target?

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Question: Is there evidence that the new protocol produces different mean HbA1c than the target?

```r
# Data summary
n <- 20
xbar <- 7.42
s <- 0.95
mu0 <- 7.0

# Test statistic
t_stat <- (xbar - mu0) / (s / sqrt(n))
cat("Test statistic: t =", round(t_stat, 3), "\n")
```

```
Test statistic: t = 1.977
```

```r
cat("Degrees of freedom:", n - 1, "\n")
```

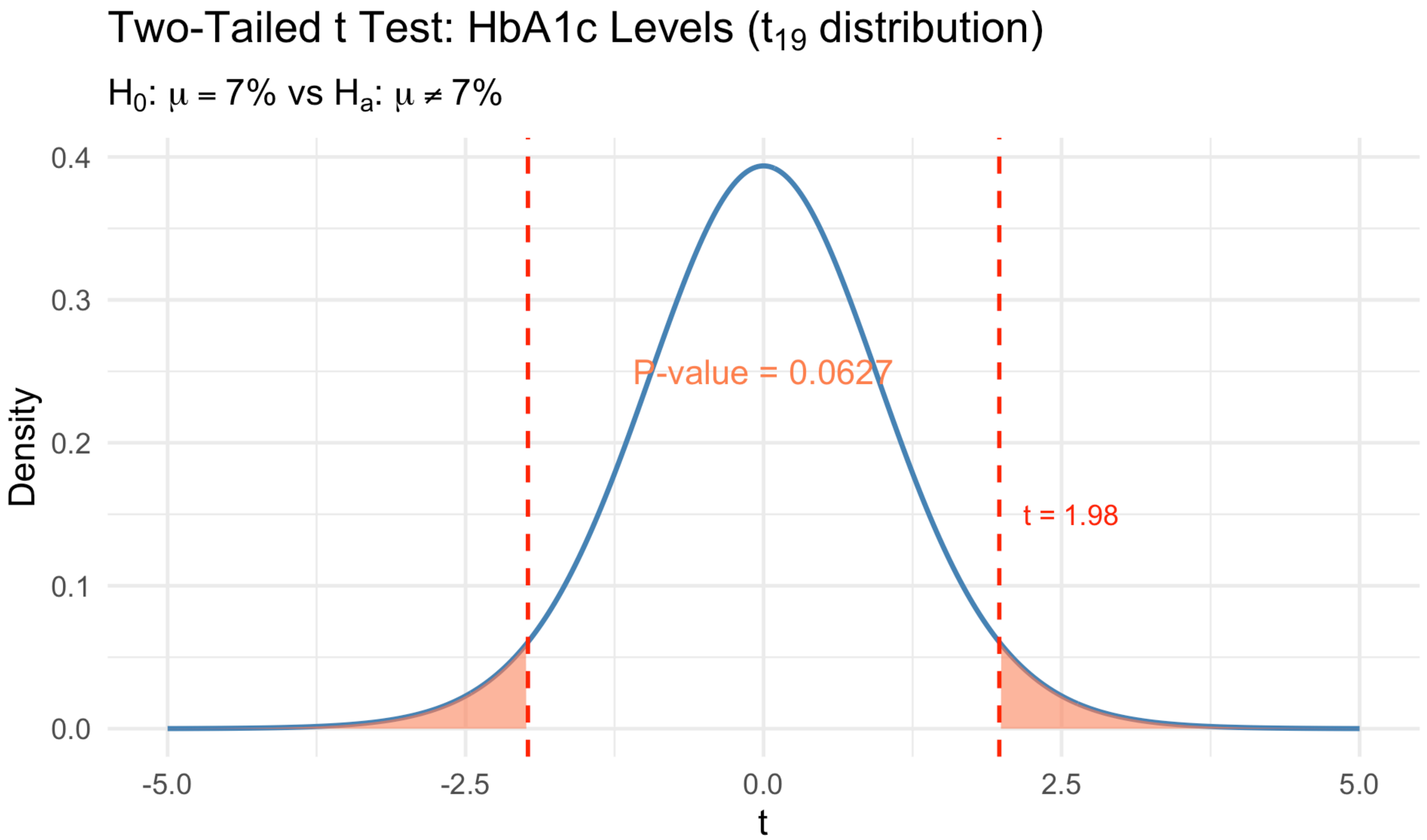
```
Degrees of freedom: 19
```

```r
# P-value (two-tailed)
p_value <- 2 * pt(abs(t_stat), df = n - 1, lower.tail = FALSE)
cat("P-value (two-tailed):", round(p_value, 4), "\n")
```

```
P-value (two-tailed): 0.0627
```

HbA1c Example: Visualizing the P-Value

▶ Code



HbA1c Example: Complete Analysis with `t.test()`

```r
# Simulate individual patient data consistent with our summary statistics
set.seed(2026)
hba1c <- rnorm(20, mean = 7.42, sd = 0.95)
# Adjust to match desired summary statistics closely
hba1c <- (hba1c - mean(hba1c)) / sd(hba1c) * 0.95 + 7.42

# Full t.test output
hba1c_test <- t.test(hba1c, mu = 7.0, alternative = "two.sided")
hba1c_test
```

```

	One Sample t-test

data:  hba1c
t = 1.9772, df = 19, p-value = 0.06272
alternative hypothesis: true mean is not equal to 7
95 percent confidence interval:
 6.975386 7.864614
sample estimates:
mean of x 
     7.42 
```

HbA1c Example: Complete Analysis with `t.test()`

```r
# Simulate individual patient data consistent with our summary statistics
set.seed(2026)
hba1c <- rnorm(20, mean = 7.42, sd = 0.95)
# Adjust to match desired summary statistics closely
hba1c <- (hba1c - mean(hba1c)) / sd(hba1c) * 0.95 + 7.42

# Full t.test output
hba1c_test <- t.test(hba1c, mu = 7.0, alternative = "two.sided")
hba1c_test
```

```
	One Sample t-test

data:  hba1c
t = 1.9772, df = 19, p-value = 0.06272
alternative hypothesis: true mean is not equal to 7
95 percent confidence interval:
 6.975386 7.864614
sample estimates:
mean of x 
     7.42 
```

Interpretation: At $\alpha = 0.05$, the P-value of 0.0627 is less than 0.05, so we reject H_0. There is statistically significant evidence that the mean HbA1c under the new protocol differs from 7.0%.

Example: Serum Transferrin Receptor in Pregnancy (Devore 9.4, Exercise 55)

Setting: The average serum transferrin receptor concentration for all women is known to be 5.63 mg/L. A study reported P-value > 0.10 for testing $H_0 : \mu = 5.63$ vs $H_a : \mu \neq 5.63$ based on $n = 176$ pregnant women.

Example: Serum Transferrin Receptor in Pregnancy (Devore 9.4, Exercise 55)

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Question: Using $\alpha = 0.01$, what would you conclude?

Conclusion: There is insufficient evidence at the 0.01 significance level to conclude that the mean serum transferrin receptor concentration for pregnant women differs from the value of 5.63 mg/L for all women.

Example: Post-Surgical Recovery Time

Setting: The standard mean recovery time after a particular surgery is 14 days. A surgeon has adopted a new minimally invasive technique and records recovery times for 15 patients.

```r
recovery <- c(11.2, 13.5, 10.8, 12.1, 14.3, 9.7, 11.9, 12.8,
              10.5, 13.1, 11.4, 12.6, 10.2, 13.8, 11.1)

# One-sided test: H0: mu = 14 vs Ha: mu < 14
recovery_test <- t.test(recovery, mu = 14, alternative = "less")
recovery_test
```

```

	One Sample t-test

data:  recovery
t = -5.7665, df = 14, p-value = 2.441e-05
alternative hypothesis: true mean is less than 14
95 percent confidence interval:
     -Inf 12.56457
sample estimates:
mean of x
 11.93333
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mean of x
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```r
cat("P-value =", format(recovery_test$p.value, scientific = TRUE), "\n\n")
```

```
P-value = 2.440503e-05
```

The Distribution of P-Values (Devore Example 9.22)

P-Values Are Random Variables

The P-value is computed from the test statistic, which is computed from sample data. Different samples yield different test statistics and therefore **different P-values**.

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Key theoretical results:

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- When H_0 is **false**: the P-value distribution is **concentrated near 0**

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Why Uniform Under H_0?

If H_0 is true, then $P(\text{P-value} \leq \alpha) = \alpha$ for any $\alpha \in [0, 1]$. This is precisely the definition of a Uniform(0, 1) distribution!

Recap: P-Value Uniformity via the Probability Integral Transform

Recall our earlier result: if T has continuous CDF F_T under H_0, then $P = 1 - F_T(T) \sim \mathrm{Uniform}(0, 1)$.

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For the z test specifically:

Let $Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$ under H_0. The P-value for an upper-tailed test is:

$$P = 1 - \Phi(Z)$$

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Since Φ is the CDF of Z, this is exactly the survival function form, and the probability integral transformation gives us $P \sim \text{Uniform}(0,1)$.

What Happens Under the Alternative?

Under $H_a : \mu > \mu_0$, the test statistic $Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$ is **not** standard normal.

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Consequence for the P-value:

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When Z tends to be large and positive, $\Phi(Z) \to 1$, so $P \to 0$.

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When Z tends to be large and positive, $\Phi(Z) \to 1$, so $P \to 0$.

The farther μ is from μ_0:

- The more Z is shifted to the right
- The larger $\Phi(Z)$ becomes
- The smaller the P-value $P = 1 - \Phi(Z)$
- The more P-values concentrate near 0

The Distribution of P-Values: A Unified View

Let $\delta = \frac{\mu - \mu_0}{\sigma/\sqrt{n}}$ be the **noncentrality parameter** (standardized effect size).

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- $\delta = 0$ (i.e., H_0 true): $P \sim \mathrm{Uniform}(0, 1)$
- $\delta > 0$ (upper-tailed H_a true): P has CDF $F_P(p) = 1 - \Phi(\Phi^{-1}(1 - p) - \delta)$

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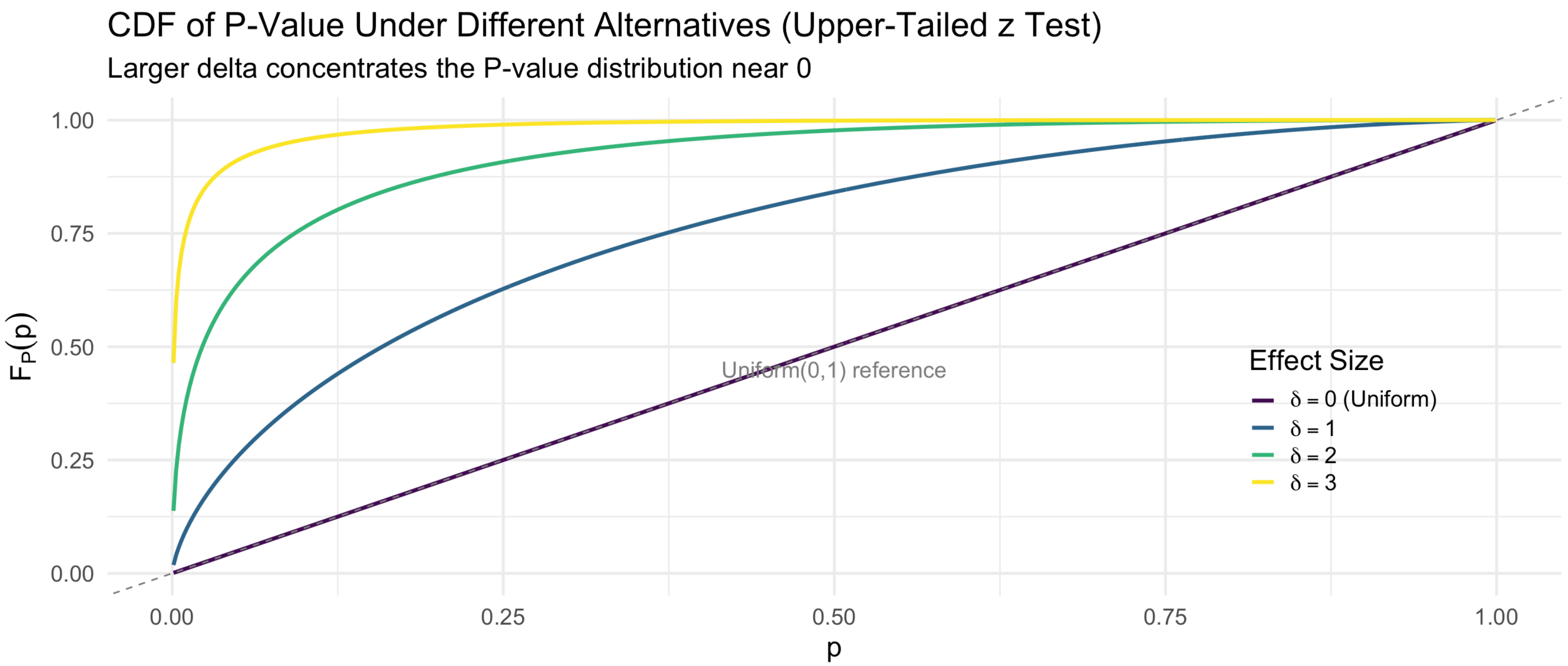
▶ Code



Power in Terms of P-Value Distribution

The **power** at a given alternative μ is the probability of rejecting H_0:

$$\text{Power}(\mu) = P_\mu(\text{Reject } H_0) = P_\mu(P \leq \alpha) = F_P(\alpha)$$

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Using our formula for the P-value CDF under the alternative:

$$\text{Power}(\mu) = 1 - \Phi\left(\Phi^{-1}(1-\alpha) - \delta\right) = \Phi\left(\delta - z_{1-\alpha}\right)$$

where $\delta = \frac{\mu - \mu_0}{\sigma/\sqrt{n}}$ and $z_{1-\alpha} = \Phi^{-1}(1-\alpha)$.

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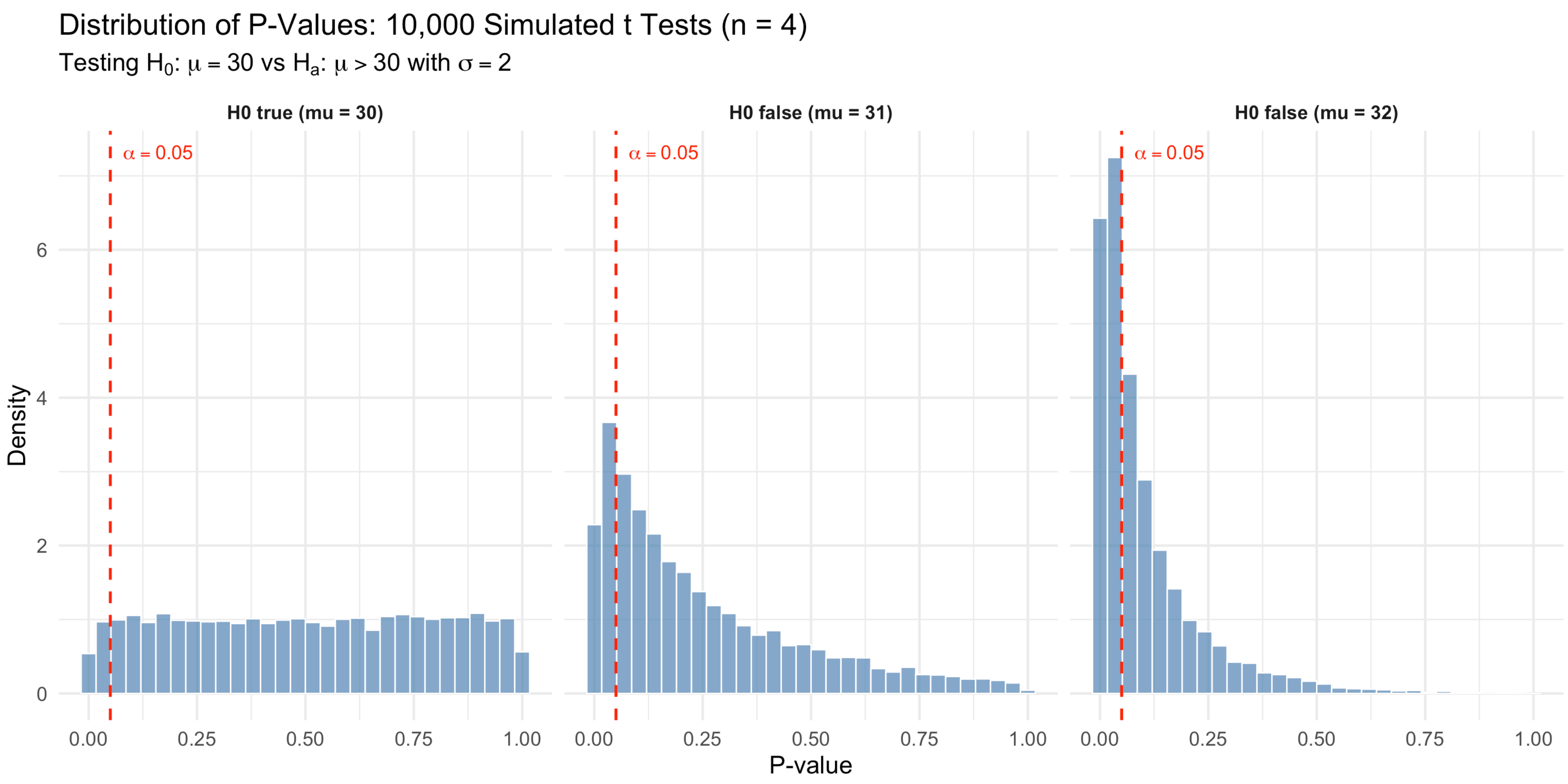
This connects P-value theory to power calculations!

Simulation: P-Value Distribution (Devore Example 9.22)

▶ Code



What the Simulation Shows

```r
# Proportion of P-values <= 0.05 in each scenario
cat("Proportion rejecting H0 at alpha = 0.05:\n")
```

```
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```

```r
cat("  H0 true (mu = 30): ", round(mean(pvals_H0_true <= 0.05), 3),
    " (should be ~0.05)\n")
```

```
  H0 true (mu = 30):  0.051  (should be ~0.05)
```

```r
cat("  H0 false (mu = 31):", round(mean(pvals_mu31 <= 0.05), 3),
    " (power at mu = 31)\n")
```

```
  H0 false (mu = 31): 0.201  (power at mu = 31)
```

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cat("  H0 false (mu = 32):", round(mean(pvals_mu32 <= 0.05), 3),
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```

```
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```

Key insights (Devore Example 9.22):

- When H_0 is true, the P-value histogram is approximately **flat** (Uniform)
- Only ~5% of P-values fall below 0.05, confirming α controls the Type I error rate
- As μ moves farther from $\mu_0 = 30$, P-values concentrate closer to 0
- Even at $\mu = 32$, power is less than 50% because $n = 4$ is very small!

Common Misinterpretations of P-Values

What the P-Value is NOT

1. **Not** the probability that H_0 is true: P-value $\neq P(H_0 \text{ is true} \mid \text{data})$
2. **Not** the probability of making an error: it is $P(\text{data this extreme} \mid H_0 \text{ true})$
3. **Not** a measure of effect size: a small P-value can arise from a tiny effect with large n
4. A P-value of 0.05 does **not** mean there is a 5% chance H_0 is true
5. Failing to reject H_0 does **not** mean H_0 is true (absence of evidence $\neq$ evidence of absence)

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ASA Statement on P-Values (2016)

The American Statistical Association emphasized: "a P-value does not provide a good measure of evidence regarding a model or hypothesis." P-values should be supplemented with effect sizes, confidence intervals, and clinical judgment.

Statistical vs Clinical Significance

▶ Code

```
n = 5000
```

▶ Code

```
Sample mean: 120.37 mmHg
```

▶ Code

```
P-value: 0.083
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▶ Code

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95% CI: ( 119.95 , 120.8 )
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▶ Code

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Statistically significant (P < 0.05)?  FALSE
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Power for the t Test: The Noncentral t Distribution

Review: Power for the z Test (Lesson 13)

For the one-sample z test (σ known), we derived closed-form expressions:

Alternative	$\beta(\mu')$
$H_a : \mu > \mu_0$	$\Phi\left(z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right)$
$H_a : \mu < \mu_0$	$1 - \Phi\left(-z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right)$
$H_a : \mu \neq \mu_0$	$\Phi\left(z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right) - \Phi\left(-z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right)$

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$H_a : \mu \neq \mu_0$	$\Phi\left(z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right) - \Phi\left(-z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right)$

Limitation: These formulas require σ to be known. In practice, we use the t test, and the analysis of power becomes more complex.

The Noncentral t Distribution (Devore 9.2)

When H_0 is **false** (i.e., $\mu = \mu' \neq \mu_0$), the t test statistic

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

does **not** follow a t_{n-1} distribution. Instead, it follows a **noncentral t distribution**.

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Definition (Devore 9.2)

Let $Z \sim N(0,1)$ and $Y \sim \chi^2_m$ be independent. For any real number δ, the random variable

$$\frac{Z + \delta}{\sqrt{Y/m}}$$

has a **noncentral t distribution** with m degrees of freedom and **noncentrality parameter** δ.

When $\delta = 0$, this reduces to the ordinary (central) t_m distribution.

Noncentrality Parameter for the One-Sample t Test

When $\mu = \mu'$ (the true mean under H_a), the one-sample t statistic has a noncentral t distribution with:

- **Degrees of freedom:** $n - 1$
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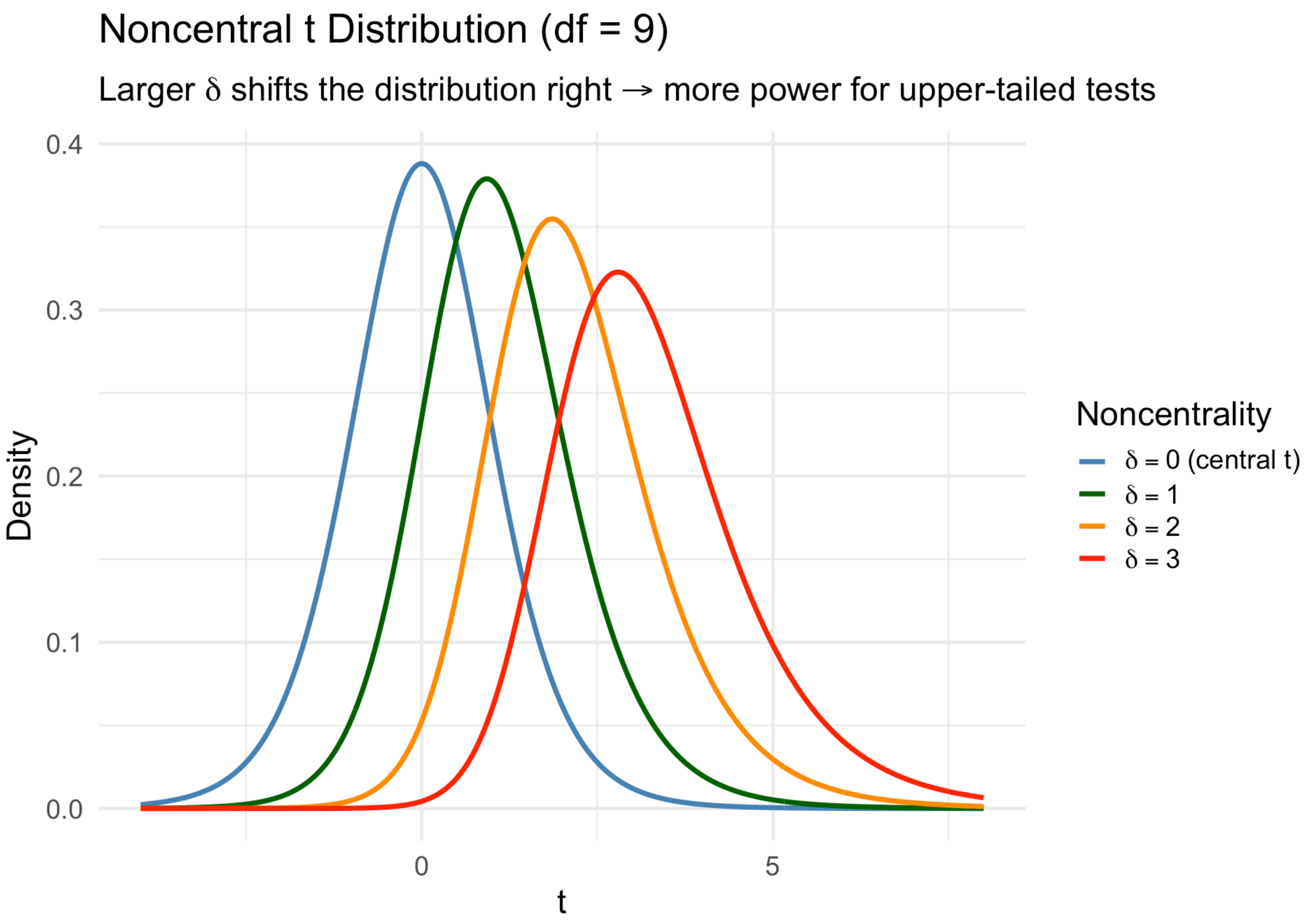
The noncentrality parameter δ measures how far the true mean is from the null value, in units of the standard error.

```r
# Example: mu_0 = 120, mu' = 125, sigma = 15, n = 25
mu0 <- 120; mu_prime <- 125; sigma <- 15; n <- 25
delta <- (mu_prime - mu0) / (sigma / sqrt(n))
cat("Noncentrality parameter: delta =", round(delta, 3))
```

```
Noncentrality parameter: delta = 1.667
```

Visualizing the Noncentral t Distribution

▶ Code



Computing Power with the Noncentral t

Power for an upper-tailed test $H_a: \mu > \mu_0$:

$$\text{Power} = P(T \geq t_{\alpha,n-1}) = 1 - F(t_{\alpha,n-1};\, n-1,\, \delta)$$

where $F(x; m, \delta)$ is the CDF of the noncentral t distribution.

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In R: `pt(t_crit, df, ncp = delta, lower.tail = FALSE)`

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In R: `pt(t_crit, df, ncp = delta, lower.tail = FALSE)`

```r
# Example (Devore Example 9.14 adapted to medical context):
# Testing whether mean systolic BP exceeds 120 mmHg
# H0: mu = 120, Ha: mu > 120, alpha = 0.05, n = 10, sigma = 10
# What is the power when mu' = 125?
n <- 10; mu0 <- 120; mu_prime <- 125; sigma <- 10; alpha <- 0.05

delta <- (mu_prime - mu0) / (sigma / sqrt(n))
t_crit <- qt(1 - alpha, df = n - 1)

power <- pt(t_crit, df = n - 1, ncp = delta, lower.tail = FALSE)
cat("Noncentrality parameter: delta =", round(delta, 3), "\n")
```

```
Noncentrality parameter: delta = 1.581
```

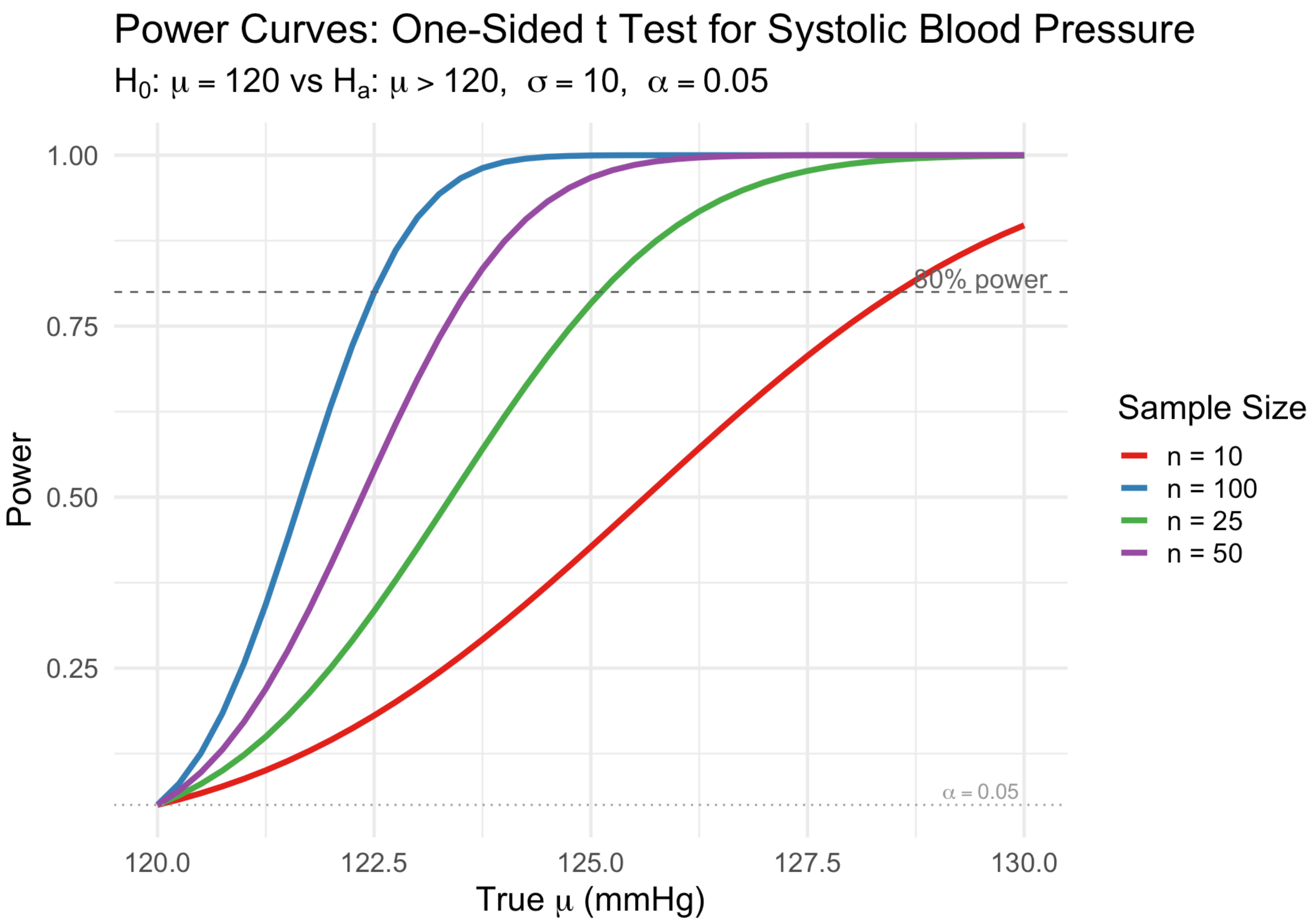
```r
cat("Critical value: t_{0.05, 9} =", round(t_crit, 3), "\n")
```

```
Critical value: t_{0.05, 9} = 1.833
```

```r
cat("Power =", round(power, 4), "\n")
```

Power Curves for the t Test: Medical Example

▶ Code



Key Features of Power Curves

Two intuitive features:

1. For any **fixed** departure from H_0, power **increases** with sample size
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Factors that increase power:

- **Larger** n: more data $\Rightarrow$ smaller SE $\Rightarrow$ easier to detect departures
- **Larger effect size** $|\mu' - \mu_0|$: bigger departures are easier to detect
- **Smaller** σ: less noise $\Rightarrow$ easier to detect signal
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Clinical Trial Implication

The power analysis must be conducted **before** collecting data. It ensures the study has sufficient sample size to detect a **clinically meaningful** effect.

Sample Size Determination

Why Determine Sample Size?

Clinical scenario: You are designing a randomized trial to test whether a new antihypertensive drug lowers systolic blood pressure more than a placebo.

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> **The Three Ingredients for Sample Size**
>
> 1. **Significance level** α (usually 0.05)
> 2. **Desired power** $1 - \beta$ (usually 0.80 or 0.90)
> 3. **Clinically meaningful effect size** $\Delta = |\mu' - \mu_0|$ (determined by clinical judgment)
> 4. **(Estimated) standard deviation** σ

Sample Size Formula for the z Test (Review)

For the one-sample z test, the required sample size for power $1-\beta$ at alternative μ':

$$n = \begin{cases} \left[\dfrac{\sigma(z_\alpha + z_\beta)}{\mu_0 - \mu'}\right]^2 & \text{one-tailed test} \\[2em] \left[\dfrac{\sigma(z_{\alpha/2} + z_\beta)}{\mu_0 - \mu'}\right]^2 & \text{two-tailed test (approximate)} \end{cases}$$

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```r
# How many patients to detect a 5 mmHg reduction with 80% power?
delta <- 5  # clinically meaningful difference (mmHg)
sigma <- 15  # estimated SD
alpha <- 0.05
beta <- 0.20  # power = 0.80

# Two-tailed z test
n_z <- (sigma * (qnorm(1 - alpha/2) + qnorm(1 - beta)) / delta)^2
cat("z test sample size (two-tailed):", ceiling(n_z), "patients")
```

```
z test sample size (two-tailed): 71 patients
```

R's `power.t.test()` Function

R provides `power.t.test()` for sample size and power calculations for the one-sample and two-sample t tests. It accounts for the noncentral t distribution automatically.

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```r
# Same example using power.t.test()
result <- power.t.test(
  delta = 5,             # difference to detect
  sd = 15,               # estimated standard deviation
  sig.level = 0.05,      # alpha
  power = 0.80,          # desired power
  type = "one.sample",  # one-sample test
  alternative = "two.sided"
)
result
```

```
     One-sample t test power calculation 

              n = 72.58407
          delta = 5
             sd = 15
      sig.level = 0.05
          power = 0.8
    alternative = two.sided
```

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  sd = 15,                # estimated standard deviation
  sig.level = 0.05,       # alpha
  power = 0.80,           # desired power
  type = "one.sample",  # one-sample test
  alternative = "two.sided"
)
result
```

```
     One-sample t test power calculation 

              n = 72.58407
          delta = 5
             sd = 15
      sig.level = 0.05
          power = 0.8
    alternative = two.sided
```

```r
cat("Required sample size:", ceiling(result$n), "patients")
```

Example: Planning a Clinical Trial for Blood Pressure

Setting: Researchers are planning a trial to test whether a new drug reduces systolic blood pressure. Historical data suggests $\sigma \approx 12$ mmHg. A reduction of 5 mmHg is considered clinically meaningful.

```r
# One-sided test: H0: mu >= 0 vs Ha: mu < 0 (reduction in BP)
power_result <- power.t.test(
  delta = 5,
  sd = 12,
  sig.level = 0.05,
  power = 0.80,
  type = "one.sample",
  alternative = "one.sided"
)
cat("For 80% power:", ceiling(power_result$n), "patients needed\n")
```

```
For 80% power: 38 patients needed
```

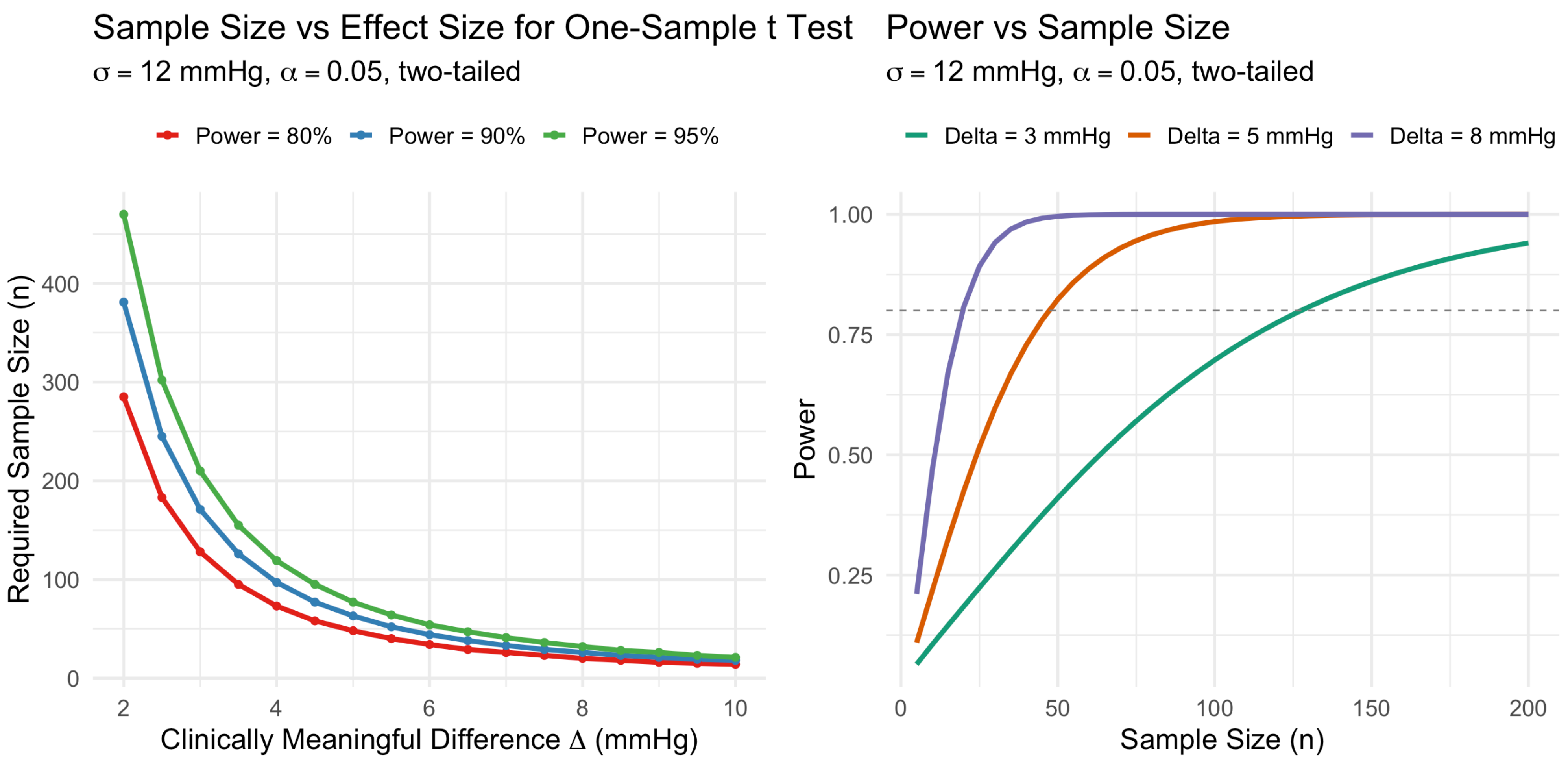
```r
# What about 90% power?
power_90 <- power.t.test(
  delta = 5, sd = 12, sig.level = 0.05,
  power = 0.90, type = "one.sample", alternative = "one.sided"
)
cat("For 90% power:", ceiling(power_90$n), "patients needed\n")
```

```
For 90% power: 51 patients needed
```

```r
# What about a smaller effect (3 mmHg)?
```

Sample Size as a Function of Effect Size and Power

▶ Code



Example: Neonatal ICU Glucose Monitoring

Setting: A neonatal ICU has historically maintained mean blood glucose at 80 mg/dL (sd = 20 mg/dL) in premature infants. A new feeding protocol is introduced. The clinical team wants to detect any shift in mean glucose of at least 8 mg/dL with 90% power.

```r
# How many infants are needed?
nicu_power <- power.t.test(
  delta = 8,
  sd = 20,
  sig.level = 0.05,
  power = 0.90,
  type = "one.sample",
  alternative = "two.sided"
)

cat("=== NICU Study Design ===\n")
```

```
=== NICU Study Design ===
```

```r
cat("Effect to detect: 8 mg/dL\n")
```

```
Effect to detect: 8 mg/dL
```

```r
cat("Assumed SD: 20 mg/dL\n")
```

```
Assumed SD: 20 mg/dL
```

```r
cat("Significance level: 0.05\n")
```

Manual Power Calculation with the Noncentral t

Let's verify `power.t.test()` using the noncentral t distribution directly:

```r
# NICU example: n = 70, delta = 8, sigma = 20, alpha = 0.05, two-tailed
n <- ceiling(nicu_power$n)
mu0 <- 80; mu_prime <- 88; sigma <- 20
alpha <- 0.05

# Noncentrality parameter
delta_ncp <- (mu_prime - mu0) / (sigma / sqrt(n))
cat("Noncentrality parameter: delta =", round(delta_ncp, 3), "\n")
```

```
Noncentrality parameter: delta = 3.298
```

```r
# Critical values for two-tailed test
t_crit <- qt(1 - alpha/2, df = n - 1)
cat("Critical value: t_{0.025,", n-1, "} =", round(t_crit, 3), "\n")
```

```
Critical value: t_{0.025, 67 } = 1.996
```

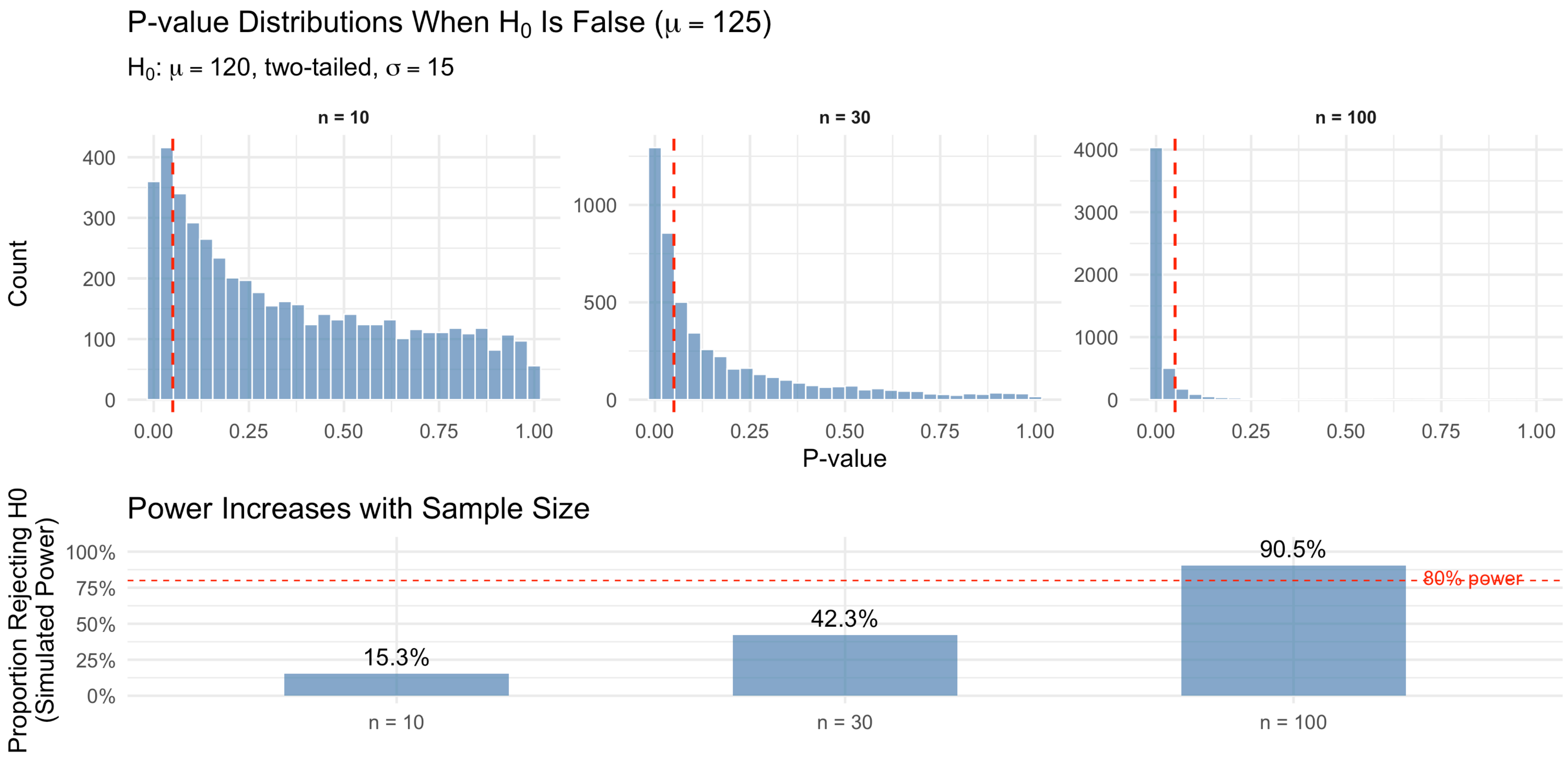
```r
# Power = P(|T| >= t_crit) under noncentral t
power_upper <- pt(t_crit, df = n - 1, ncp = delta_ncp, lower.tail = FALSE)
power_lower <- pt(-t_crit, df = n - 1, ncp = delta_ncp, lower.tail = TRUE)
power_total <- power_upper + power_lower

cat("Power (upper tail):", round(power_upper, 4), "\n")
```

```
Power (upper tail): 0.9016
```

Connecting P-Values, Power, and Sample Size

The Complete Picture

▶ Code



Summary: P-Values and Their Relationship to Testing

Concept	What It Tells You
P-value	Strength of evidence against H_0 (smaller = stronger)
α	Threshold for "significant" evidence (set in advance)
$\beta(\mu')$	Probability of missing a true effect of size μ'
Power $= 1 - \beta$	Probability of detecting a true effect
Sample size n	Number of observations needed for desired power

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The chain of reasoning:

$$\text{Larger } n \implies \text{Smaller SE} \implies \text{Larger } |T| \implies \text{Smaller P-values} \implies \text{More Power}$$

Practical Guidelines for Medical Research

Best Practices

1. **Always report P-values AND confidence intervals** — the CI tells you about effect size
2. **Conduct power analysis BEFORE the study** — not after!
3. **Use clinically meaningful effect sizes** — not just "any" difference
4. **Don't p-hack:** pre-register hypotheses and analysis plans
5. **Interpret in context:** a P-value of 0.048 vs 0.052 should not lead to radically different conclusions
6. **Consider multiple comparisons:** if testing many hypotheses, adjust α (Bonferroni, FDR, etc.)

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FDA Guidance

Clinical trials typically require $\alpha = 0.05$ (two-sided) and power ≥ 0.80. Phase III trials often require power ≥ 0.90 to ensure adequate evidence for drug approval.

Multiple Testing and P-Value Adjustment

The Multiple Testing Problem

Scenario: A genomics study tests 20,000 genes for differential expression between tumor and normal tissue. Using $\alpha = 0.05$ for each test...

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> ⚠ **The Core Problem**
>
> When conducting m hypothesis tests, the probability of making **at least one** Type I error grows rapidly with m, even if each individual test has size α.

Family-Wise Error Rate (FWER)

Definition

The **family-wise error rate (FWER)** is the probability of making one or more Type I errors among all m tests:

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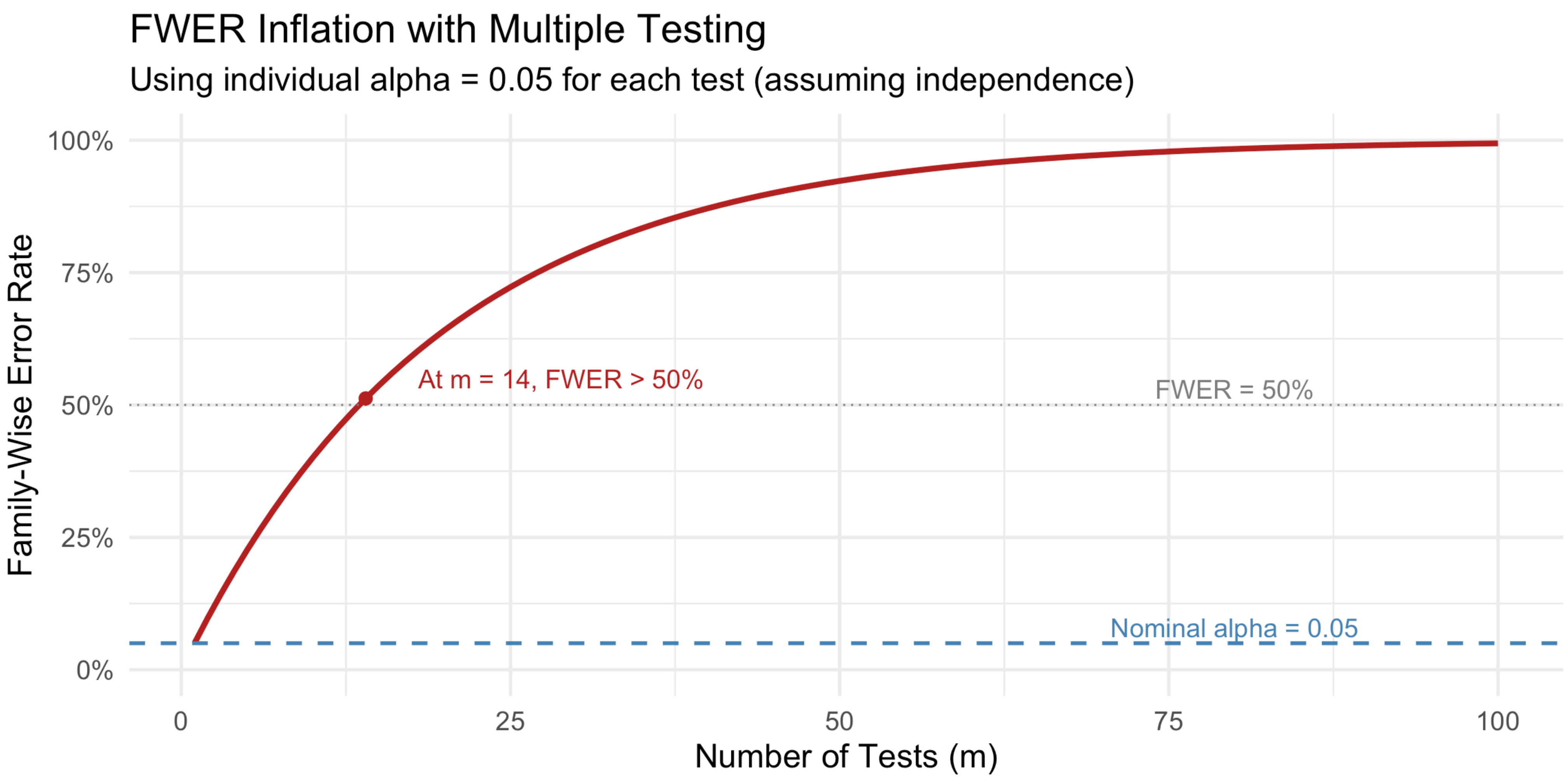
```r
# FWER as a function of m for alpha = 0.05
m_tests <- c(1, 2, 5, 10, 20, 50, 100)
fwer <- 1 - (1 - 0.05)^m_tests
tibble(m = m_tests, FWER = round(fwer, 3)) |>
  pivot_wider(names_from = m, values_from = FWER) |>
  knitr::kable(col.names = paste0("m = ", m_tests))
```

m = 1	m = 2	m = 5	m = 10	m = 20	m = 50	m = 100
0.05	0.098	0.226	0.401	0.642	0.923	0.994

Visualizing the FWER Inflation

▶ Code



The Bonferroni Correction

Bonferroni Correction

To control the FWER at level α when conducting m tests, reject each individual hypothesis only if its P-value satisfies:

$$p_i \leq \frac{\alpha}{m}$$

Equivalently, compute **adjusted P-values**: $\tilde{p}_i = \min(m \cdot p_i, 1)$

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Theoretical justification (Boole's inequality):

$$\text{FWER} = P\left(\bigcup_{i \in \mathcal{H}_0} \{\text{Reject } H_{0i}\}\right) \leq \sum_{i \in \mathcal{H}_0} P(\text{Reject } H_{0i}) \leq m_0 \cdot \frac{\alpha}{m} \leq \alpha$$

where m_0 is the number of true null hypotheses.

Bonferroni: Example

Clinical trial: Testing a drug's effect on 5 different biomarkers.

```r
# Raw P-values from 5 tests
biomarkers <- c("CRP", "IL-6", "TNF-alpha", "Cortisol", "Fibrinogen")
raw_pvalues <- c(0.008, 0.023, 0.041, 0.067, 0.12)

# Bonferroni adjustment
m <- length(raw_pvalues)
bonf_adjusted <- pmin(raw_pvalues * m, 1)  # p.adjust does this

# Using R's built-in function
bonf_adjusted_r <- p.adjust(raw_pvalues, method = "bonferroni")

tibble(
  Biomarker = biomarkers,
  `Raw P-value` = raw_pvalues,
  `Bonferroni Adjusted` = round(bonf_adjusted_r, 3),
  `Significant (alpha=0.05)` = ifelse(bonf_adjusted_r <= 0.05, "Yes", "No")
) |> knitr::kable()
```

Biomarker	Raw P-value	Bonferroni Adjusted	Significant (alpha=0.05)
CRP	0.008	0.040	Yes
IL-6	0.023	0.115	No

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Limitations of Bonferroni

Bonferroni is Conservative

The Bonferroni correction controls FWER at level **at most** α, but often the actual FWER is much smaller, leading to reduced **power**.

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When Bonferroni is too conservative:

- Tests are positively correlated (common in genomics)
- Large number of tests ($m > 100$)
- Many true alternative hypotheses

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- Large number of tests ($m > 100$)
- Many true alternative hypotheses

The problem: With $m = 20{,}000$ tests, Bonferroni requires $p < 0.05/20{,}000 = 2.5 \times 10^{-6}$. This may miss many true effects.

Alternative: False Discovery Rate (FDR)

Instead of controlling the probability of **any** false positives (FWER), we can control the **proportion** of false positives among all discoveries.

Definition: False Discovery Rate

Let V = number of false discoveries (Type I errors) and R = total number of rejections. The **false discovery rate** is:

$$\text{FDR} = E\left[\frac{V}{R} \mid R > 0\right] \cdot P(R > 0)$$

Often approximated as $E[V/R]$ with the convention that $V/R = 0$ when $R = 0$.

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Interpretation: FDR = 0.05 means that, on average, 5% of our "discoveries" are false positives. This is often acceptable in exploratory research.

Benjamini-Hochberg (BH) Procedure

Benjamini-Hochberg Procedure (1995)

To control FDR at level q:

1. Order the m P-values: $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$
2. Find the largest k such that $p_{(k)} \leq \frac{k}{m} q$
3. Reject all hypotheses with $p_{(i)} \leq p_{(k)}$ (i.e., reject $H_{(1)}, \ldots, H_{(k)}$)

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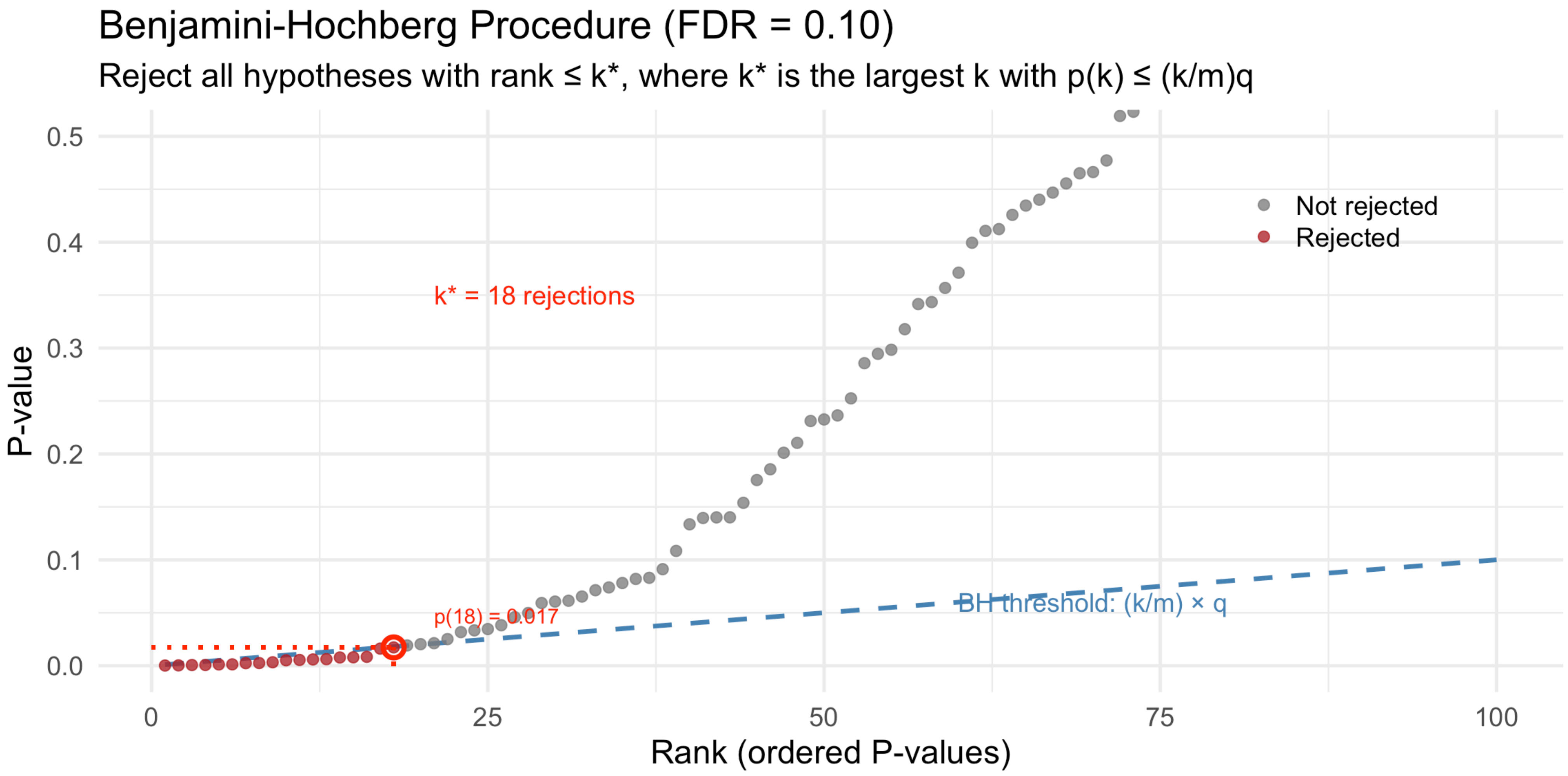
Adjusted P-values: The BH-adjusted P-value for the i-th ordered hypothesis is:

$$\tilde{p}_{(i)} = \min_{j \geq i} \left\{ \frac{m}{j} p_{(j)} \right\}$$

(enforcing monotonicity: larger raw P-values cannot have smaller adjusted P-values)

Visualizing the BH Procedure

▶ Code



BH Procedure: Example

```r
# Same biomarker example
biomarkers <- c("CRP", "IL-6", "TNF-alpha", "Cortisol", "Fibrinogen")
raw_pvalues <- c(0.008, 0.023, 0.041, 0.067, 0.12)

# Multiple adjustment methods
adjustments <- tibble(
  Biomarker = biomarkers,
  `Raw P` = raw_pvalues,
  Bonferroni = p.adjust(raw_pvalues, method = "bonferroni"),
  Holm = p.adjust(raw_pvalues, method = "holm"),
  BH = p.adjust(raw_pvalues, method = "BH"),
  BY = p.adjust(raw_pvalues, method = "BY")
)

adjustments |>
  mutate(across(where(is.numeric), ~round(., 3))) |>
  knitr::kable()
```

Biomarker	Raw P	Bonferroni	Holm	BH	BY
CRP	0.008	0.040	0.040	0.040	0.091
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Comparison of Adjustment Methods

Method	Controls	Formula	Conservative?
Bonferroni	FWER	$\tilde{p}_i = m \cdot p_i$	Very
Holm	FWER	Step-down Bonferroni	Less than Bonferroni
Hochberg	FWER	Step-up Bonferroni	Less than Holm
BH	FDR	$\tilde{p}_{(i)} = \min_{j \geq i} \frac{m}{j} p_{(j)}$	Not very
BY	FDR (any dependence)	BH with correction factor	More than BH

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Practical Guidance

- Use **Bonferroni** or **Holm** when controlling FWER is essential (confirmatory studies, regulatory submissions)
- Use **BH** for exploratory research where some false discoveries are acceptable (genomics, screening)
- Use **BY** when tests may have arbitrary dependence structure

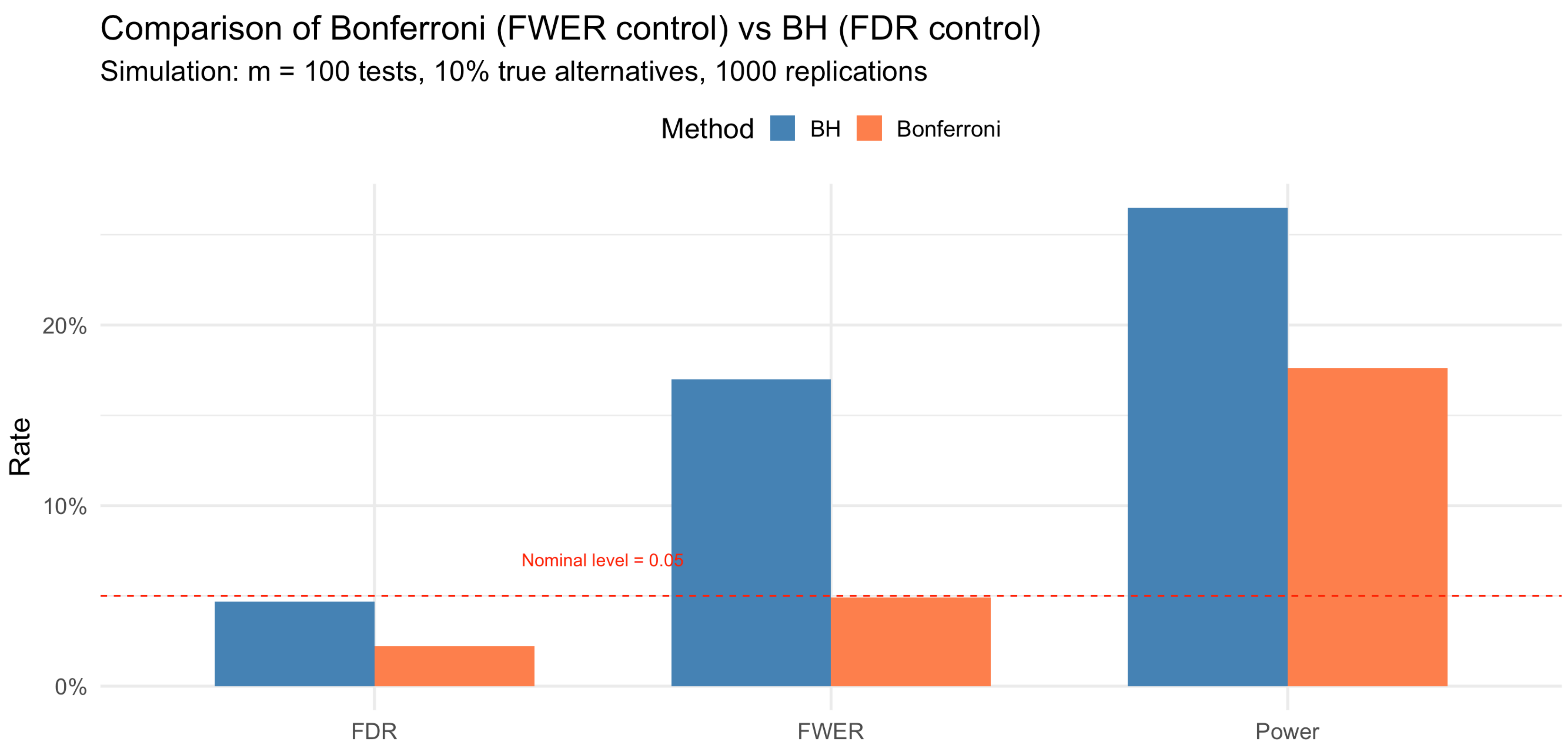
Multiple Testing in R: `p.adjust()`

```r
# Simulate a genomics-style analysis
set.seed(551)
m <- 1000  # 1000 genes
n_true_effects <- 50  # 50 genes truly differentially expressed

# Generate P-values
pvals <- c(
  rbeta(n_true_effects, 0.5, 10),  # True effects: small P-values
  runif(m - n_true_effects)         # Null: uniform P-values
)

# Apply different correction methods
results <- tibble(
  raw = pvals,
  bonferroni = p.adjust(pvals, method = "bonferroni"),
  holm = p.adjust(pvals, method = "holm"),
  BH = p.adjust(pvals, method = "BH")
)

# Count discoveries at alpha/q = 0.05
cat("Number of 'significant' results (adjusted P <= 0.05):\n")
```

```
Number of 'significant' results (adjusted P <= 0.05):
```

Simulation: Comparing FWER and FDR Control

▶ Code



When to Use Which Correction?

Decision Framework

Use FWER control (Bonferroni/Holm) when:

- False positives have serious consequences (e.g., drug approval)
- Small number of pre-specified hypotheses
- Confirmatory analysis with regulatory implications

Use FDR control (BH) when:

- Exploratory analysis where findings will be validated
- Large number of tests (genomics, neuroimaging)
- Some false discoveries are acceptable in exchange for more power
- Generating hypotheses for future studies

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Medical research examples:

- **Phase III clinical trial** (primary endpoint): Bonferroni or no adjustment (single test)
- **Genome-wide association study**: BH or even less stringent (will be replicated)
- **Multiple secondary endpoints**: Holm or Hochberg
- **Biomarker discovery panel**: BH

Multiple Testing: Summary

1. **The problem:** Multiple tests inflate the chance of false discoveries
2. **FWER** = P(at least one false positive); controlled by **Bonferroni**, **Holm**
3. **FDR** = E(proportion of false discoveries); controlled by **BH**, **BY**
4. **Trade-off:** Stricter error control $\Rightarrow$ lower power
5. **In R:** `p.adjust(pvalues, method = "bonferroni")` or `method = "BH"`

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5. **In R:** `p.adjust(pvalues, method = "bonferroni")` or `method = "BH"`

Key Insight

Multiple testing adjustment is not optional—it's a fundamental requirement for honest science. Always report whether and how P-values were adjusted.

Key Takeaways and Looking Ahead

Key Takeaways

Today's Main Points

1. **P-value definition:** probability of observing data as extreme or more extreme, assuming H_0 is true
2. **P-value uniformity:** Under H_0, P-values are Uniform(0,1) by the probability integral transformation
3. **Size of a test:** P-value tests have exact size α because $P(P \leq \alpha) = \alpha$ when $P \sim \text{Uniform}(0, 1)$
4. **Conditional P-values:** Fisher's exact test conditions on sufficient statistics to eliminate nuisance parameters
5. **Noncentral t:** when H_0 is false, T follows a noncentral t with $\delta = (\mu' - \mu_0)/(\sigma/\sqrt{n})$
6. **Power in R:** `pt(t_crit, df, ncp = delta)` for manual; `power.t.test()` for convenience
7. **Multiple testing:** FWER controlled by Bonferroni/Holm; FDR controlled by BH procedure

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7. **Multiple testing:** FWER controlled by Bonferroni/Holm; FDR controlled by BH procedure

Common mistakes to avoid:

- Interpreting P-value as $P(H_0 \text{ true})$
- Confusing statistical and clinical significance
- Computing power **after** seeing the data (post hoc power)
- Failing to adjust for multiple comparisons in exploratory analyses
- Using Bonferroni when FDR control would be more appropriate (and vice versa)

Looking Ahead

Next class (Lesson 15): Neyman-Pearson Lemma; Likelihood Ratio Tests (Devore 9.5)

- Most powerful tests for simple hypotheses
- The Neyman-Pearson framework
- Likelihood ratio test statistic
- Connection to tests we've already learned

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Sections 9.2, 9.4
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Sections 8.1–8.2
- Wasserstein, R. L., & Lazar, N. A. (2016). The ASA Statement on p-Values: Context, Process, and Purpose. *The American Statistician*, 70(2), 129–133.
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- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.

Questions?

Thank you!